

Mata Kuliah - Kode Mata Kuliah	Pembelajaran Mesin - 52311
Hari, tanggal - Kelas	Jumat, 24 April 2026 – CS6
Waktu - Sifat	5 hari kerja – Deadline Senin, 4 Mei 2026
Dosen/Koordinator Mata Kuliah	Dr. Ariana Yunita



Segala bentuk kecurangan akademik (termasuk pelanggaran tata tertib ujian) dalam setiap proses pembelajaran akan diberikan sanksi sesuai peraturan yang berlaku



CASE STUDY

(CPMK-1) (Bobot : 50%)

(CPMK-2) (Bobot : 50%)

The Free Nutritious Meals program (Makan Bergizi Gratis/MBG) has raised concerns in Indonesia, particularly regarding budget allocation, food types, and distribution logistics. A critical issue worth highlighting from a nutritional standpoint is the uniform standardization of food composition across all students, even though each individual has distinct nutritional requirements. A one-size-fits-all approach to meal portioning should not be applied to the MBG program. Nutritional personalization is one viable solution to address this issue, as it ensures that the dietary needs of each student receiving MBG are adequately met.

However, personalization introduces its own challenge, as it adds considerable complexity for MBG program administrators. According to several studies in literature, this challenge can be addressed through clustering, whereby students are grouped based on their nutritional needs. Such groupings would assist MBG administrators in delivering appropriate meal portions and nutritional content to each group.

As a pilot project, synthetic data was generated to identify the existing nutritional clusters. **Your role is a data engineer, and the objective of this case is to build a machine learning classification model that can categorize students into nutritional groups, so that each student's dietary needs can be properly and effectively fulfilled. Propose and evaluate the model.**

Report: A report should be no more than 6 pages. Write your report documenting the entire process and evaluation. There is no prescribed template; you are encouraged to present your report creatively. Submit your report in PDF format along with your source code. Your report must include the preprocessing steps and evaluation results. Additionally, justify your choices of method, the baseline model, and the evaluation metrics used.

